Before you tier memory over routable PCIe, answer four questions

For HPC and platform teams with idle memory on other nodes. General questions, no benchmarks.

Routable PCIe sits between RDMA and CXL on latency

Ways to reach memory beyond the local socket, slowest first

01 **RDMA**

Reads and writes remote memory through the network card.

02 **Routable PCIe**

Maps a window of another host's memory across a PCIe fabric, without cache coherency.

03 **CXL**

Memory the CPU reaches with plain loads and stores, kept cache coherent.

Question 1: who owns a page when caches are not coherent?

01

The risk

Without coherency, two hosts can hold different copies of the same data. Hardware will not reconcile them.

02

One way round it

Give each page a single owner at a time, as swap does.

Question 2: which pages are worth sending to another node?

01 **The target**

Pages too cold to keep in local DRAM and too hot to push out to disk.

02 **Measure first**

Log how often each page is touched before you move anything.

Question 3: how much does lending slow down the lender?

01

Shared links

Borrowed pages travel over the lending node's PCIe links and through its memory.

02

What to track

The lending node's own job speed, with and without a borrower.

Question 4: where do pages go when the lender wants its memory back?

01 **Reclaim**

The lender's own job can grow at any time. Borrowed pages then need another home.

02 **Plan the exit**

Decide up front whether they move to another node or fall back to disk.

Tiering over routable PCIe is still an open research question

> **Tiering memory over routable PCIe has barely been studied, so the first useful output is a measurement.**

Separately, our research collaboration with a Norwegian research laboratory is designing dSwap, a swap tier over routable PCIe. In progress, no results yet.

What would you move into a borrowed memory tier first?

Scoping is short, fixed and ends in a written plan.

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